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THE PATENT ACT 1970
(39 of 1970)

The Patents Rules, 2003 PROVISIONAL/COUP
 LET SPECIFICATION
 (See section 10 and rule 13)

1. TITLE OF THE INVENTION Smart Grid Based Prepaid Metering System	
2. APPLICANT (1) KCG College of Technology	
3. PREAMBLE TO THE DESCRIPTION	
PROVISIONAL	COMPLETE The following specification particularly describes the invention and the manner in which it is to be performed.
4. DESCRIPTION (Description shall start from next page.)	
5. CLAIMS (not applicable for provisional specification. Claims should start with the preamble— "I/we claim" on separate page)	
1. The Smart Grid Based Prepaid Metering System uses cutting-edge communication module consisting of Wi-Fi, cellular, and LoRaWAN networks. 2. The Smart Grid Based Prepaid Metering System the prepaid credit management system further comprises an authentication module to verify prepaid transactions and prevent fraud. 3. The Smart Grid Based Prepaid Metering System the central server includes a data analytics module configured to analyze consumption patterns and predict future usage. 4. The Smart Grid Based Prepaid Metering System provides notifications to users via a mobile application when the prepaid credits fall below a predetermined threshold. 5. The Smart Grid Based Prepaid Metering System includes a mobile application for users to view and manage their prepaid balance and energy consumption. 6. The Smart Grid Based Prepaid Metering System comprising a backup power supply unit integrated with the smart meter to ensure continued operation during power outages. 7. The Smart Grid Based Prepaid Metering System includes a local display for users to view real-time consumption and credit balance.	

8.The Smart Grid Based Prepaid Metering System is configured to send automated notifications to users via SMS or email when their prepaid credit balance is low.

9.The Smart Grid Based Prepaid Metering System the prepaid credit management unit includes a module for tiered pricing based on consumption levels.

10.The Smart Grid Based Prepaid Metering System is further configured to support integration with third-party energy management systems for enhanced monitoring and optimization.

6.DATEANDSIGNATURE(tobegivenattheendoflastpageofspecification)

7. **ABSTRACT**OF TNE INVENTION(lo be given along with complete specificationon separatepage)

The present invention relates to a Smart Grid Based Prepaid Metering System designed to integrate with modern smart grid infrastructures for efficient management of electrical consumption and billing. The system comprises a smart meter equipped with real-time consumption measurement and a communication module for data exchange with a central server. The central server manages prepaid credit balances for multiple smart meters, updates credit information, and monitors energy usage patterns. The prepaid credit management unit within each smart meter handles credit storage, deduction based on consumption, and generates alerts when the balance is low. Users and utility providers can access a user interface, including mobile applications and web portals, to manage credits, view usage data, and

receive notifications. The system enhances energy management by providing real-time feedback, preventing over-consumption, and enabling flexible billing options. It supports various communication networks and includes features such as authentication, security, and predictive analytics for optimized energy use and maintenance.

Note:-

*Repeatboxesin caseofmorethanoneentry.

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pplicantsshouldbe giveninfull,**family**nameInthebeginning.

'Con.pleteaddressof**theapplicant**shouldbegivenstatingthepostalindexno./code,ctateand
country.

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